

E078

Calculus II for Engineers

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

An engineer modeling heat buildup in a battery pack, working against a tight deadline, needs $\int x e^x dx$. The pack is rated at 75 kWh. Which antiderivative is correct?

- A. $(x - 1)e^x + C$
- B. $(x^2/2)e^x + C$
- C. $x e^x + C$
- D. $(x + 1)e^x + C$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A signal-processing student in a crowded campus lab evaluates $\int 2x \cos(x^2) dx$ for a filter design due Friday. Which result is correct?

- A. $x^2 \sin(x^2) + C$
- B. $-\sin(x^2) + C$
- C. $2x \sin(x^2) + C$
- D. $\sin(x^2) + C$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A water plant's flow sensor records 4, 6, and 10 liters per hour at $t = 0$, 1, and 2 hours. The sensor was calibrated by a technician last Tuesday, and the plant serves 12,000 homes. Using the trapezoidal rule, about how many liters flowed from $t = 0$ to $t = 2$?

- A. 13 liters
- B. 26 liters
- C. 16 liters
- D. 10 liters

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A steel-alloy damper in a bridge, installed by a contractor from another state, swings 10 cm on its first swing, and each later swing is 60% of the one before. About what total distance do all the swings add up to?

- A. Infinite, because there are infinitely many swings.
- B. 25 cm
- C. About 16.7 cm
- D. 6 cm

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

An infinite geometric series has ratio 0.75 and sums to 40. What is its first term?

- A. 40
- B. 160
- C. 30
- D. 10

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A numerical-methods team at an aerospace supplier with 300 employees finds that its error terms follow $e_n = 1/n^{1.5}$. Their lead, who prefers whiteboards to slides, asks whether the total error $\sum e_n$ is finite. Which conclusion and reason are correct?

- A. It diverges, like the harmonic series.
- B. It diverges, because p must exceed 2.
- C. It converges, because the terms approach 0.
- D. It converges, because it is a p -series with $p = 1.5 > 1$.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A student claims: “The terms of $\sum 1/n$ approach 0, so the series converges.” Which response is correct?

- A.** The claim is right; any series whose terms go to 0 converges.
- B.** The claim is wrong; the harmonic series diverges, so terms going to 0 does not guarantee convergence.
- C.** The claim is right, because it is a geometric series with $r < 1$.
- D.** The claim is wrong, because the terms do not approach 0.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A cloud provider housed in a cold northern climate finds that stored data grows at a rate proportional to the amount stored: $dy/dt = 0.1y$, with t in months. It starts at 50 TB. The provider recently changed its logo. About how much data is stored after 10 months?

- A.** About 18 TB
- B.** 100 TB
- C.** About 136 TB
- D.** 150 TB

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A quantity follows $dy/dt = ky$ and doubles every 7 years. What is k ?

- A.** About 0.286 per year
- B.** About 0.099 per year
- C.** About 0.143 per year
- D.** About 10.1 per year

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

An orange robot arm in a warehouse that runs 24 hours a day moves along $x = t^2$, $y = t^3$. What is the slope dy/dx of its path at $t = 2$?

- A.** 12
- B.** $1/3$
- C.** 3
- D.** 2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A radar console built by a well-known defense contractor reports a target at $r = 4$ km, $\theta = 60^\circ$. The operator has 15 years of experience. What are the target's Cartesian coordinates?

- A. $(2, 2\sqrt{3})$ km
- B. $(4, 60)$ km
- C. $(0.5, 0.87)$ km
- D. $(2\sqrt{3}, 2)$ km

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

Convert the Cartesian point $(0, -3)$ to polar form with $r > 0$ and $0 \leq \theta < 2\pi$.

- A. $(3, \pi/2)$
- B. $(3, \pi)$
- C. $(3, 3\pi/2)$
- D. $(9, 3\pi/2)$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

An engineer modeling heat buildup in a battery pack, ~~working against a tight deadline,~~ needs $\int x e^x dx$. ~~The pack is rated at 75 kWh.~~ Which antiderivative is correct?

Target word: antiderivative of a product **Decides:** By parts with $u = x$, $dv = e^x dx$: $x e^x - \int e^x dx$.

A	Correct	$x e^x - e^x + C = (x - 1)e^x + C$.
B	Adjacent category	Integrates each factor separately; products do not integrate that way.
C	Dropped element	Stops after uv and drops the $-\int v du$ term.
D	Inverted operation	Adds the remaining integral instead of subtracting it.

2. Answer D Concept 2

A signal-processing student ~~in a crowded campus lab~~ evaluates $\int 2x \cos(x^2) dx$ for a filter design ~~due Friday~~. Which result is correct?

Target word: antiderivative using substitution **Decides:** $u = x^2$, $du = 2x dx$, so the integral is $\int \cos u du$.

A	Adjacent category	Integrates $2x$ and $\cos(x^2)$ separately and multiplies.
B	Inverted operation	Uses the derivative sign of cosine instead of its antiderivative.
C	Dropped element	Keeps the $2x$ factor even though it was absorbed into du .
D	Correct	$\int \cos u du = \sin u + C = \sin(x^2) + C$.

3. Answer A Concept 3

A water plant's flow sensor records 4, 6, and 10 liters per hour at $t = 0$, 1, and 2 hours. ~~The sensor was calibrated by a technician last Tuesday, and the plant serves 12,000 homes.~~ Using the trapezoidal rule, about how many liters flowed from $t = 0$ to $t = 2$?

Target word: trapezoidal estimate of total flow **Decides:** $h = 1$; $T = (1/2)[4 + 2(6) + 10]$.

A	Correct	$(1/2)(4 + 12 + 10) = 13$.
B	Dropped element	Leaves out the $h/2$ factor.
C	Confusion set	Right Riemann sum $(6 + 10)$, a neighboring method, not the trapezoidal rule.
D	Adjacent category	Left Riemann sum $(4 + 6)$, a different method.

4. Answer B Concept 4

A ~~steel alloy~~ damper in a bridge, ~~installed by a contractor from another state,~~ swings 10 cm on its first swing, and each later swing is 60% of the one before. About what total distance do all the swings add up to?

Target word: sum of an infinite geometric series **Decides:** $a = 10$, $r = 0.6$, $|r| < 1$: $\text{sum} = 10 / (1 - 0.6)$.

A	Absolute language	A geometric series with $ r < 1$ has a finite sum.
B	Correct	$10 / 0.4 = 25$.
C	Inverted operation	Divides by r (0.6) instead of $1 - r$.
D	Wrong stem number	Only the second swing (10×0.6) .

5. Answer D Concept 4 · transfer

An infinite geometric series has ratio 0.75 and sums to 40. What is its first term?

Target word: solve for the first term **Decides:** $a = S(1 - r) = 40 \times 0.25$.

A	Wrong stem number	The sum, not the first term.
B	Inverted operation	Divides the sum by $1 - r$ instead of multiplying.
C	Confusion set	Multiplies by r instead of $1 - r$.
D	Correct	$40 \times (1 - 0.75) = 10$.

6. Answer D Concept 5

A numerical-methods team at an aerospace supplier ~~with 300 employees~~ finds that its error terms follow $e_n = 1/n^{1.5}$. Their lead, ~~who prefers whiteboards to slides~~, asks whether the total error $\sum e_n$ is finite. Which conclusion and reason are correct?

Target word: convergence with correct reason **Decides:** p-series with $p = 1.5 > 1$.

A	Confusion set	The harmonic series has $p = 1$; here $p = 1.5$.
B	Wrong stem number	The cutoff is $p > 1$, not $p > 2$.
C	Inverted operation	Terms approaching 0 is necessary, not sufficient; the reason is invalid.
D	Correct	p-series converge exactly when $p > 1$.

7. Answer B Concept 5 · transfer

A student claims: “The terms of $\sum 1/n$ approach 0, so the series converges.” Which response is correct?

Target word: evaluate the reasoning **Decides:** Harmonic series ($p = 1$) diverges even though its terms go to 0.

A	Absolute language	$\sum 1/n$ is the classic counterexample.
B	Correct	The divergence test can only rule convergence out, never in.
C	Adjacent category	$\sum 1/n$ is not geometric; it has no constant ratio.
D	Dropped element	$1/n$ does approach 0; the error is in the conclusion drawn.

8. Answer C Concept 6

A cloud provider ~~housed in a cold northern climate~~ finds that stored data grows at a rate proportional to the amount stored: $dy/dt = 0.1y$, with t in months. It starts at 50 TB. ~~The provider recently changed its logo.~~ About how much data is stored after 10 months?

Target word: solve the separable equation **Decides:** $y = 50e^{\{0.1 \times 10\}} = 50e$.

A	Inverted operation	$50e^{\{-1\}}$: uses decay instead of growth.
B	Adjacent category	Linear growth: $50(1 + 0.1 \times 10)$.
C	Correct	$50 \times e^1 \approx 50 \times 2.718 \approx 136$.
D	Wrong stem number	Adds 10 months $\times$ 10 TB to 50 TB as if 0.1 meant 10 TB a month.

9. Answer B Concept 6 · transfer

A quantity follows $dy/dt = ky$ and doubles every 7 years. What is k ?

Target word: solve for the rate constant **Decides:** $2 = e^{7k}$, so $k = \ln 2 / 7$.

A	Dropped element	$2 / 7$ leaves out the natural log.
B	Correct	$0.693 / 7 \approx 0.099$.
C	Adjacent category	$1 / 7$ treats doubling as a linear 100% gain spread evenly.
D	Inverted operation	$7 / \ln 2$ flips the fraction.

10. Answer C Concept 7

An ~~orange robot arm in a warehouse that runs 24 hours a day~~ moves along $x = t^2$, $y = t^3$. What is the slope dy/dx of its path at $t = 2$?

Target word: parametric slope **Decides:** $dy/dt = 3t^2 = 12$ and $dx/dt = 2t = 4$ at $t = 2$.

A	Dropped element	Uses dy/dt alone.
B	Inverted operation	Divides dx/dt by dy/dt .
C	Correct	$12 / 4 = 3$.
D	Adjacent category	$y / x = 8 / 4$ divides positions, not rates.

11. Answer A Concept 8

A radar console ~~built by a well-known defense contractor~~ reports a target at $r = 4$ km, $\theta = 60^\circ$. ~~The operator has 15 years of experience.~~ What are the target's Cartesian coordinates?

Target word: polar to Cartesian **Decides:** $x = 4 \cos 60^\circ = 2$; $y = 4 \sin 60^\circ = 2\sqrt{3}$.

A	Correct	$4 \times 0.5 = 2$ and $4 \times (\sqrt{3}/2) = 2\sqrt{3} \approx 3.46$.
B	Confusion set	Treats the polar pair as if it were already Cartesian.
C	Dropped element	Leaves out $r = 4$.
D	Inverted operation	Swaps sine and cosine.

12. Answer C Concept 8 · transfer

Convert the Cartesian point $(0, -3)$ to polar form with $r > 0$ and $0 \leq \theta < 2\pi$.

Target word: Cartesian to polar **Decides:** $r = 3$; the point is on the negative y-axis.

A	Inverted operation	Points to the positive y-axis; ignores the sign.
B	Adjacent category	Points to the negative x-axis, the wrong axis.
C	Correct	$r = \sqrt{0^2 + 9} = 3$; the negative y-axis is at $3\pi/2$.
D	Dropped element	Forgets the square root in r .

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	4	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Integration by parts	$\int u \, dv = uv - \int v \, du$; choose u as the factor that simplifies when differentiated (e.g., the polynomial in $x \cdot e^x$).	Integrating each factor separately and multiplying the results.	A product of two unlike functions, such as a polynomial times an exponential or trig function.
2	u-substitution	When the integrand contains a function and (a constant multiple of) its derivative, set u to the inner function so the derivative becomes du .	Keeping the leftover derivative factor after it has already become du .	An inner function whose derivative also appears in the integrand.
3	Numerical integration (trapezoidal rule)	Trapezoidal estimate = $(h/2)[f_0 + 2f_1 + \dots + 2f_{n-1} + f_n]$ for equal spacing h .	Forgetting the $h/2$ factor or doubling the endpoints.	A table of measured values at equal time or distance steps.
4	Geometric series sum	$a + ar + ar^2 + \dots = a / (1 - r)$ when $ r < 1$; it diverges when $ r \geq 1$.	Dividing by r instead of $1 - r$, or assuming infinitely many terms must add to infinity.	Each term is a fixed fraction of the one before.
5	Convergence tests (p-series and divergence test)	$\sum 1/n^p$ converges only when $p > 1$; terms going to 0 is necessary for convergence but not sufficient.	Concluding a series converges because its terms approach 0.	A series whose terms shrink toward zero and a question about its sum.
6	Separable differential equations: exponential models	$dy/dt = ky$ separates to $y = y_0 e^{kt}$; doubling time = $\ln 2 / k$.	Treating proportional growth as linear growth.	"Rate proportional to the current amount."
7	Derivatives of parametric curves	$dy/dx = (dy/dt) / (dx/dt)$, evaluated at the given t .	Using dy/dt alone, or dividing position values instead of rates.	x and y each given as functions of a parameter t .
8	Polar coordinates	$x = r \cos \theta$, $y = r \sin \theta$; $r = \sqrt{x^2 + y^2}$ and θ is set by the point's quadrant.	Swapping sine and cosine, or ignoring the quadrant.	A point given as (r, θ) or a conversion between systems.

Course scope (public description): This course advances learners' understanding of calculus with a focus on integration techniques, sequences and series, differential equations, and alternative coordinate systems. Learners apply advanced integration and numerical approximation methods to model and solve engineering and computational problems, analyze infinite processes using sequences, series, and convergence tests, and represent dynamic systems through separable differential equations, parametric equations, and polar coordinates. Emphasizing real-world applications, the course highlights the role of calculus in optimization, numerical methods, and systematic problem-solving across engineering disciplines.